Reg. No. | | | | | | | | | | |

[12ME53/13ME53]

FIFTH SEMESTER B.E/B.TECH EXAMINATION, DECEMBER-2015

MANUFACTURING PROCESS AND SYSTEMS-II

Time: 3 Hours **Maximum Marks:70**

Instructions:

Answer any **FIVE** full Questions.

1. a. With a suitable sketch , explain the nomenclature of a single point cutting tool **9**

b. In an orthogonal cutting process, following data were observed: chip length of 80 mm was obtained with an uncut chip length of 200 mm and the rake angle of 20° and depth of cut 0.5 mm. The horizontal and vertical components of cutting force F_H & F_V were 2000 N and 200 N respectively. Determine the shear plane angle. Chip thickness, friction angle and resultant cutting force. **5**

2. a. List and explain the essential properties of cutting tool materials **8**

b. Write short notes on following cutting tool materials **6**

a. High speed steels (HSS) b. Cemented Carbides

3. a. Write the differences between a capstan lathe and a turret lathe. **8**

b. A 50 mm diameter C.I workpiece of 200 mm length is rough turned on lathe with a tool of lead angle of 20° using a cutting speed of 70 m/min, federate 0.4 mm/rev, depth of cut as 4mm. the specific cutting energy for material is 0.07kw/cm^3/min. Determine material removal rate, power and torque requirement of spindle, main cutting force, machining time, uncut chip thickness . **6**

4. a. Write the differences between a planning machine and a shaping machine . **8**

b. Explain with suitable sketches following drilling operations **6**

i. Reaming

ii. Boring

iii. Tapping

5. a. Give a detailed classification of milling machines **5**

b. What is indexing? Name different methods of indexing. Briefly explain compound indexing method. **9**

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6.	**a.**	Explain the factors to be considered in selecting a grinding wheel.	**8**
	b.	List and explain different abrasive materials used in grinding wheel.	**6**
7.	**a.**	Give a detailed classification of broaching machines	**6**
	b.	What is Lapping? What are different types of lapping operations? List the advantages & disadvantages of lapping	**8**
8.	**a.**	With a suitable sketch, explain ultra-sonic machining (USM)	**9**
	b.	Discuss the effects of various cutting parameters on metal removal rate in Abrasive Jet Machining (AJM).	**5**